

Exam 1

MATH 2025

Instructions

- Exam length: 75 minutes. Complete the exam in class using handwritten work only. Show supporting reasoning where appropriate.
- You may use a calculator and one sheet of handwritten notes (front and back).
- Read each prompt carefully. If you believe a question is ambiguous, ask Dr. Friedlander.
- Unless a computation is explicitly requested, focus on conceptual explanations.
- When code or output is supplied, interpret it as directed.
- Clearly label each answer. Partial credit may be awarded for well-communicated reasoning.

The exam centers on data collected from a community rail trail. The data set contains information from 90 different days which are the rows in the data set. We will focus on the following variables from the dataset:

- **volume**: records the number of users counted on the trail that day
- **hightemp**: day's high temperature (°F)

Summary information and figures from the data and model are provided at the end of this exam. Use Tables 1-3 and Figures 1-3 as needed; you do not need to reproduce them.

Section A. Multiple Choice

Circle the single best answer for each question. Each question is worth 4 points.

1. Which step of the data analysis cycle is most closely associated with evaluating whether the simple linear regression conditions appear reasonable for this rail trail model?
 - (A) Pose a question
 - (B) Obtain data
 - (C) Understand the data
 - (D) Communicate findings
2. In Figure 1, what do the scatterplot and regression line suggest about the association between **volume** and **hightemp**?
 - (A) A strong negative linear relationship
 - (B) A moderate positive linear relationship
 - (C) No apparent relationship
 - (D) A nonlinear relationship with a clear curve
3. Which code segment would create a scatterplot of trail volume versus high temperature using `gf_point()` from the `ggformula` package?
 - (A) `gf_point(volume ~ hightemp, data = rail_trail)`
 - (B) `gf_point(hightemp, volume, data = rail_trail)`
 - (C) `gf_point(rail_trail, volume, hightemp)`
 - (D) `gf_point(~ volume + hightemp, data = rail_trail)`
4. Figure 3 contains a null distribution for the slope of a regression model predicting **volume** from **hightemp**. Complete this sentence. The randomization distribution was most likely generated by:
 - (A) Repeatedly shuffling **hightemp** to mimic the effect of measurement error in the predictor
 - (B) Repeatedly shuffling **volume**, to simulate the null hypothesis of no relationship
 - (C) Repeatedly resampling (bootstrapping) the data with replacement to estimate variability in the slope
 - (D) Repeatedly shuffling both variables simultaneously, to preserve the correlation structure

5. Compared with a 95% confidence interval for the mean trail volume when `hightemp` = 80°F, a 95% prediction interval for a single day at 80°F will be:
- (A) Narrower, because individual days are less variable than the regression line
 - (B) Wider, because it must account for both model uncertainty and day-to-day variability
 - (C) The same width, because they are based on the same fitted regression line
 - (D) Uninterpretable without a theoretical t distribution assumption
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Section B. Short Answer

Show all reasoning. Each question is worth 8 points.

1. **Checking conditions:** List all condition for inference of a linear regression model. Refer to Figure 2. For each condition, note whether you can assess the condition from this plot? If you can, briefly note what you observe. If you can't, briefly describe how you would assess it.

2. **Using model output:** Based on Table 2, write the fitted regression equation and explain what the slope means in context. Then state the predicted trail volume on a day with a high temperature of 75°F (you will need to do a little math here).

3. **Confidence interval and hypothesis test reasoning:** Compare the differences and similarities of the null distribution (e.g. Figure 3), used while performing a hypothesis test, and the sampling distribution, used to create a confidence interval?

Section C. True/False

For each statement, circle True or False and give a *brief* explanation. Each question is worth 5 points.

1. **True / False:** If we swapped the roles of `volume` and `hightemp` in the regression, the value of the correlation between the two variables would stay the same but the slope would change.
2. **True / False:** A small p-value from the randomization test proves that high temperatures cause increased trail usage.
3. **True / False:** If more observations are added that follow the same linear trend, we would expect the standard error of the slope to decrease.
4. **True / False:** Even when the equal variance condition is satisfied, prediction intervals get wider as you move further from the mean of `hightemp`.
5. **True / False:** When the confidence interval for the slope contains zero, the corresponding hypothesis test at the same confidence level will not reject the null hypothesis.

Section D. Evaluate a Response

Each item shows a student's written response. For each, decide whether it is *correct*, *partially correct*, or *incorrect*, and explain your reasoning in 2–3 sentences. Each question is worth 10 points.

1. Student A states, “Because the observed slope (shown as the vertical line) falls far into the tail of the randomization distribution in Figure Figure 3, the p-value must be close to 1, so there is no evidence of a relationship.” Evaluate Student A's conclusion.

2. Student B writes, “We must have made a mistake in our analysis because the intercept is negative and we can’t have negative volume.” Evaluate this interpretation using information from the dataset.
3. Student C creates a different model with the data. They fit a line showing that there is a positive correlation between the high-temperature and the number of accidents and make the following claim. “Bikers are more likely to crash on days when it is hot.” Assess this claim.

Reference Materials

Summary Tables

Table 1: Selected summaries for the rail trail dataset.

n	mean_volume	sd_volume	mean_hightemp	sd_hightemp
90	375.4	127.46	68.83	13.02

Table 2: Regression output for predicting trail volume from daily high temperature.

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-17.08	59.40	-0.29	0.77	-135.11	100.96
hightemp	5.70	0.85	6.72	0.00	4.02	7.39

Table 3: Model-level summaries for the rail trail regression.

r_squared	sigma	df_residual
0.339	104.19	88

Plots

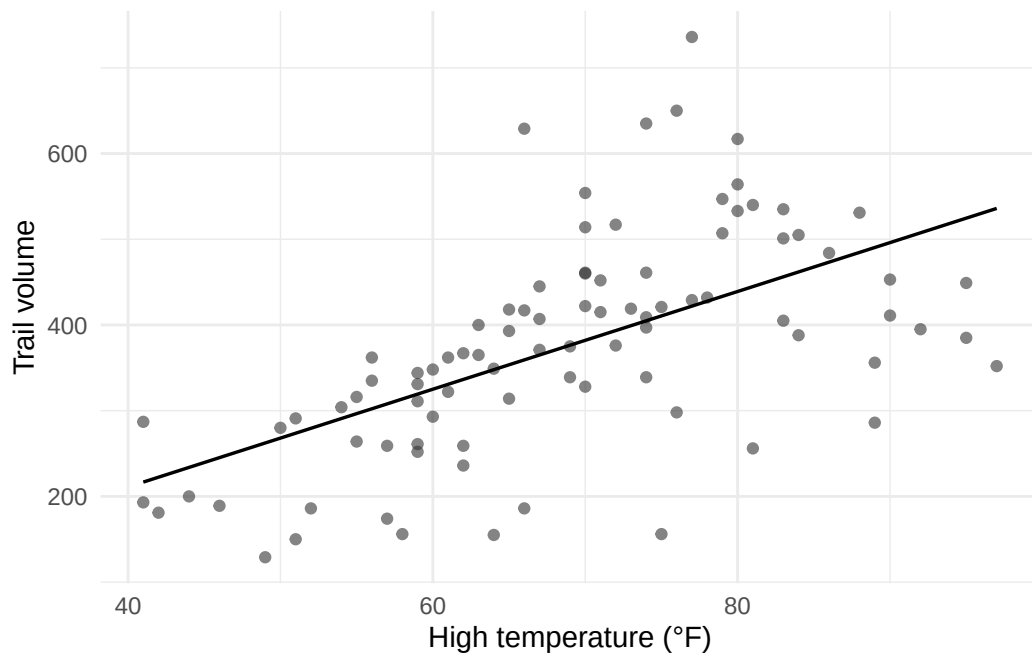


Figure 1: Trail volume vs. high temperature with fitted regression line.

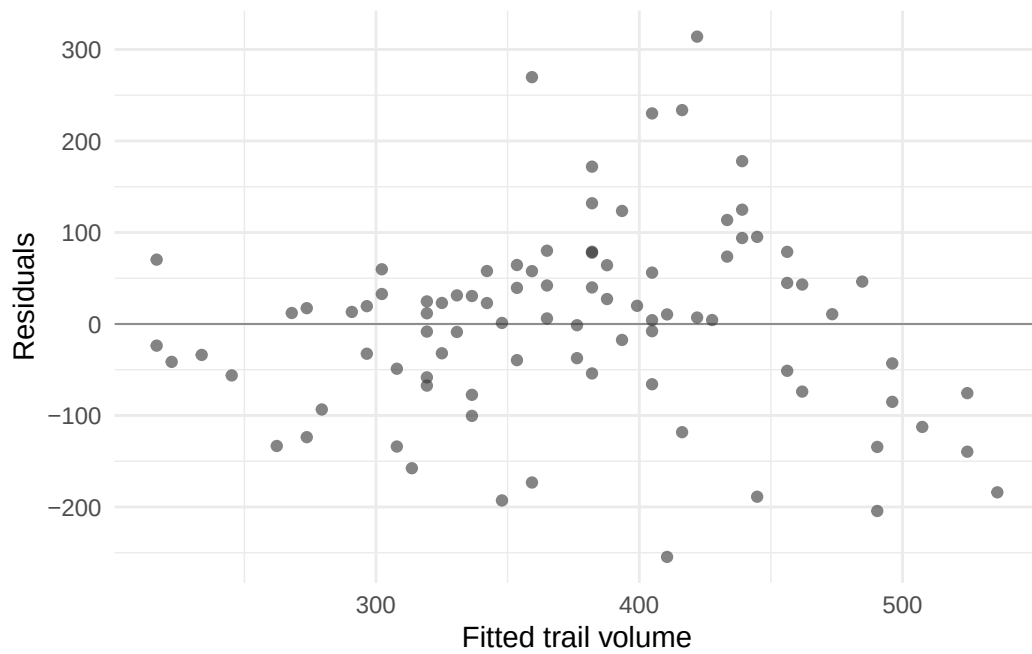


Figure 2: Residuals versus fitted values for the regression of volume on high temperature.

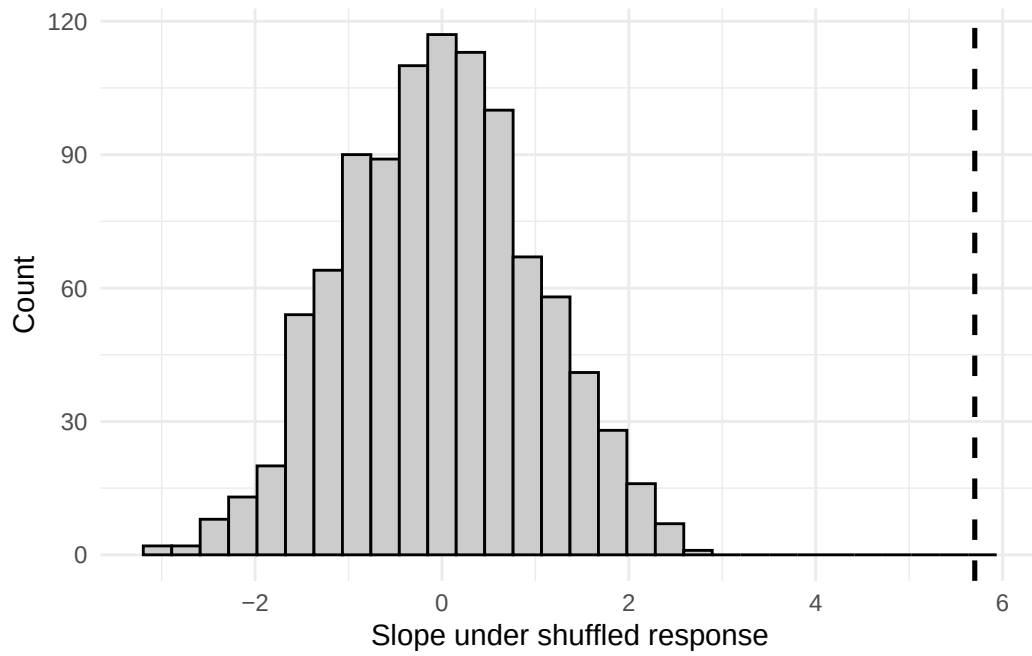


Figure 3: Randomization distribution of slope estimates assuming no relationship between temperature and trail volume.